

Following information is required for drafting of patent application.

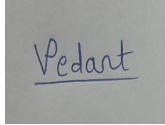
1. Full name, nationality and address of applicant(s):

Full Name	Nationality	Address
Vishwakarma Institute of Technology	Indian	666, Upper Indiranagar, Bibwewadi, Pune, Maharashtra, India – 411 037

2. Full name (including middle name), nationality, address (VIT/VIIT address), mail id, and phone number of inventor(s):

Full Name (Including middle name) (Min. Two faculty)	Nationality	VIT/VIIT Address (Start with full dept. name followed by full institute name) (For ex.)	Mail ID	Phone No.	Gender
Kaushalya Vaibhav Thopate	IN	Department of Computer Engineering, Vishwakarma Institute of Technology, Pune	kaushalya.thopte@vit.edu	9960158822	Female
Vedant Vijay Patkar	IN	Department of Electronics and Telecommunication Engineering, Vishwakarma Institute of Technology, Pune	vedant.patkar241@vit.edu	8446094203	Male

Copy and paste clear soft copy of signatures of all inventors in the following table: (Insert or delete the cells as per number of inventors)

		
Name 1	Name 2	Name 3

3. Title of the invention:

Live-Sync Digital Twin for Industrial Safety Synchronization and Context-Aware Action Denial.

4. Technical field of the invention:

The CPS and IIoT convergence brings a tidal wave of changes, including industrial security, from passive observing to active intelligent intervention. The Digital Twin (DT), a high fidelity virtual representation that is synchronized with physical assets in real time, is at the core of this invention. Instead of the traditional systems which just visualize data, the system employs Edge Computing to analyze safety critical data on site.

The primary innovation attacks "Safety Latency" the harmful time delay incurred in cloud based systems when data is whisked to a remote server and back before an emergency stop can be triggered. Locally, it runs a deterministic safety protocol on an ESP32 microcontroller to assess environmental vectors atmospheric concentrations of gases, proximity breaches, and thermal anomalies.

When a hazard is identified, the "Action Denial" logic takes precedence over any manual or remote commands and a physical hardware interlock (such as a servo driven mechanical lock) is engaged. This maintains a "Safe State" even in the event of a loss of network connectivity. Meanwhile, leveraging WebSockets for a "live sync" dashboard experience and local SD card "Black Box" forensics data redundancy, it is a robust fail functional safety ecosystem.

5. Prior art:

Traditional industrial safety systems, such as SCADA or basic PLC systems are limited by their centralized and usually reactive nature. These legacy systems function in "silos" they may sound local audible alarms or stop mechanical relays, but they do not have the communicative depth to create a high fidelity virtual mirror for a remote stakeholder. And this produces a visibility gap that prevents offsite supervisors from capturing a machine's immediate environmental context.

In addition, cloud-based IoT Digital Twins are affected by Network Jitter and Latency. In a model that depends on the cloud, a safety stop command has to go from the sensor to the cloud and back to the actuator; a delay of just a few seconds can mean the difference between a near miss and a fatal accident. This leads to "Blind Execution," machinery blindly obeys user commands in the presence of physical dangers, such as gas leaks or perimeter breaches, etc.

Your invention makes a significant contribution to the art by decentralizing safety logic to the Edge, bringing the “decision making” closer to the sensor in an ESP32 microcontroller rather than the cloud or a distant server. This change allows for a Permission Governed actuation model, where each command is checked with a local real time sensor fusion (Gas, Proximity, Temperature) prior to physical execution.

The following are examples of related investigations and projects based in industry:

- **The Azure Digital Twin Platform:** The agility provided for large-scale enterprise analytics is powerful, yet it largely depends on cloud side processing, rendering it less deterministic at the millisecond level for hardware interlocking than your approach that resides at the edge.
- **The "Safety Monitoring and Control Framework" by Zhang et al. (2019):** This project explored DT-based safety for robotic systems but primarily focused on collision detection via computer vision, whereas your system utilizes a cost-effective, multi-sensor pack (MSP) for more holistic environmental context.
- **Siemens MindSphere:** A leading industrial IoT platform that provides excellent predictive maintenance; however, it generally treats "Digital Twins" as monitoring tools rather than active safety "governors" that implement context-aware hardware action denial at the edge.

6. Object:

The present invention is aimed at overcoming the limitations of passive monitoring by creating a Live Sync Digital Twin as an active safety governor. Using a Deterministic Safety Engine at the edge, critical environmental data are locally processed on an ESP32 microcontroller. This ensures that safety critical actions, such as invoking an Action Denial mechanism, happen in a sub 100ms response time. This is a critical advance over tradition cloud-based Iot systems, as it effectively removes “Safety Latency” – the hazardous delay induced by network round trips.

When environmental parameters (for example, toxic gas levels or distance violations) are violated, the system disables machine operation an hardware interlock, so that no command manual or remote can be blindly executed. Also, high speed WebSockets keep the virtual dashboard up to date with these physical changes in the twinkle of an eye and a localized SD card based “Black Box” adds forensic redundancy. This two pronged strategy further guarantees that even in the unwelcome event of complete network outage, the system is fail functional upholding a stringent safety perimeter and maintaining an auditable log of all telemetry for the post incident forensics.

7. Synopsis:

The present invention is a complex Cyber-Physical System (CPS) based on a Multi-Sensor Pack (MSP) managed by an ESP32 microcontroller, acting as a Cyber-Physical System (CPS). This edge-computing platform combines a diverse set of sensors—ultrasonic proximity sensors, MQ series gas detectors, DHT11 thermal sensors, and Light Dependent Resistor (LDRs)—to achieve a complete “situational awareness” of the industrial scene. Instead of sending the multi-sensor fusion data to the cloud, the system locally and deterministically processes it, unlike traditional monitoring systems. This means the environmental context is assessed once every few milliseconds, creating a strong safety buffer for industrial assets that does not rely on external network conditions.

The framework also features a Hybrid Communication Scheme to reconcile the local physical action with its remote virtual representation. The system is based on high speed WebSockets (Socket.io) for the transmission of the "volatile data" to keep the React based Digital Twin dashboard updated with a latency less than 100ms. This “live sync” function lets distant supervisors instantly be crystal clear reflections of the machine state. Concurrently, the system uses RESTful APIs to interface with the persistent storage in a MongoDB cloud database and offers a local SD card "Black Box" for forensic redundancy.

The end "effect" level of operation is the "Action Denial" system which converts the Digital Twin from a passive observer into an active safety governor. Through pre defined environmental limits, the system enforces a “Permission Governed” actuation model. When a hazard is detected e.g., a worker breaches a restricted proximity zone or a surge in toxic gas the system engages a physical hardware interlock (a servo driven service lock) and a system wide lockout.

8. Brief description of drawings (if any):

Figure 1 System Architecture. This diagram illustrates the three-tier hierarchy of the invention. It shows the Physical Layer (sensors and actuators), the Communication

Layer (WebSockets for real time sync and REST for database persistence), and the Application Layer (the user dashboard and MongoDB cloud).

Figure 2 Circuit Interfacing. This schematic details the pin-level connections of the ESP32 microcontroller with the sensor suite and the hardware interlock (SG90 Servo). It highlights the path of the "Safety Interrupt" which overrides the motor control signals.

9. Detail description of the invention:

The invention is characterized by its **Action Denial Architecture**, which is divided into three functional segments:

A. The Edge Intelligence Module: The core of the system is an ESP32 microcontroller programmed with a non-blocking execution loop. It polls the sensor array every 2-5 milliseconds. Unlike standard systems that simply report data, this module acts as a "Gatekeeper." It calculates a **Safety Index** based on the fusion of gas levels and proximity. If the distance is $<15\text{cm}$ or gas concentration is $>150\text{PPM}$, the module enters a "Lockout" state.

B. The Live-Sync Communication Pipeline:

To ensure the remote supervisor sees exactly what the machine "feels," the invention uses **Socket.io**. This allows for a full-duplex, volatile data stream. When the physical interlock triggers, the virtual twin on the React dashboard changes color (e.g., from Green to Red) and disables the virtual "Start" button instantly. This prevents the "desynchronization" often seen in standard IoT platforms.

C. Forensic Data Logging (Black Box):

The system includes an SPI-based SD card module. Every 500ms, a snapshot of all sensor values and the state of the Action Denial logic is written to a local file. This provides a "ground truth" record that is immune to network outages, essential for industrial insurance and safety compliance auditing.

10. Best method of performance of the invention:

The best performance is achieved by configuring the ESP32 to handle the Servo Interlock via a Hardware Interrupt. By doing so, the mechanical lockout occurs independently of the code's communication tasks. For optimal results, the WebSocket server should be hosted on a local Edge Gateway (like a Raspberry Pi or local server) to minimize the Round Trip Time (RTT) to the dashboard, achieving a "Live-Sync"

feel where the physical and virtual states change simultaneously.

11. CLAIMS:

1. A cyber-physical safety system comprising an edge-computing microcontroller (ESP32) directly interfaced with a multi-sensor array including an ultrasonic proximity sensor and a gas concentration sensor, wherein said microcontroller executes a local deterministic logic loop to evaluate environmental safety thresholds; the system further comprises a pulse-width modulated (PWM) mechanical interlock that is physically triggered by the microcontroller to obstruct machine actuation instantly upon the detection of a threshold breach, ensuring safety enforcement is handled at the hardware level independently of any external network or cloud-based commands.
2. A real-time synchronization apparatus for industrial digital twins comprising a WebSocket communication interface established between a physical asset's sensor pack and a remote React-based virtual dashboard; the apparatus is configured to package raw environmental telemetry into volatile data packets and emit them via full-duplex channels to ensure the virtual representation reflects the physical state with a latency of less than 100 milliseconds; this ensures that the virtual twin's control interface is programmatically locked or updated in high-fidelity synchronization with the physical asset's localized safety state.
3. An industrial data logging system providing forensic redundancy for digital twins, comprising an SPI-interfaced local storage module (SD card) and a secondary RESTful API cloud-link; the system is configured to perform simultaneous data writes where high-resolution sensor telemetry and safety-state transitions are recorded to the local storage module every 500 milliseconds, while a lower-resolution summary is transmitted to a cloud database (MongoDB); this dual-pathway architecture ensures a permanent, unalterable "Black Box" record of safety events remains physically accessible on-site even in the event of total network failure or cloud disconnection.
4. A context-aware control framework for industrial machinery that implements a "pre-flight" safety validation protocol; the system comprises a processing unit that intercepts all incoming actuation signals—whether manual or remote—and compares the request against a real-time sensor-fusion safety index derived from proximity, gas, and thermal telemetry; the framework is characterized by a "Gatekeeper" logic which denies the execution of any movement command if the physical environmental parameters are outside of safe operational bounds, thereby preventing "blind

execution" of commands during hazardous conditions.

5. A safety latching mechanism for industrial equipment characterized by a persistent software-emergency state; the system is configured such that once the hardware interlock is triggered by a sensor breach, the system enters a "latched" mode that maintains the mechanical lockout even after environmental sensors return to a safe range; the system further comprises a secure reset protocol requiring a secondary authentication signal—generated either by a physical tamper-proof button or a secure cryptographic token from a remote supervisor—to disengage the lockout, ensuring human-in-the-loop verification before the industrial process can be resumed.

12. Inventive step of your invention:

The inventive step resides in the Context-Aware interlocking mechanism. Traditional twins are merely "mirrors" that observe; this invention turns the twin into a "governor." By integrating the logic that says *"If the environment is unsafe, the command is denied at the hardware level,"* the system removes the risk of human error or network delay. The transition from Passive Monitoring to Proactive Action Denial represents a significant advancement in Cyber Physical safety.

13. Industrial application:

The invention is highly applicable in:

- Robotic Manufacturing Cells: Preventing high speed robots from moving if a human worker breaches the light curtain or proximity sensor.
- Chemical Warehousing: Automatically locking doors or disabling ventilation fans if toxic gas sensors detect a leak, while providing a remote twin for emergency responders.
- Power Plants: Monitoring high temperature zones and automatically denying access/operation to equipment if thermal thresholds are exceeded.

14. Abstract:

The invention provides a Live-Sync Digital Twin for Industrial Safety utilizing a hybrid edge-cloud architecture. An ESP32 microcontroller processes a multi-sensor pack (gas, proximity, thermal, optical) to perform local, high-speed safety evaluations. It implements Action Denial by triggering physical hardware interlocks in under 100ms when hazards are detected. Simultaneously, real-time data is mirrored to a React dashboard via WebSockets for remote oversight. The system ensures a "Safe State" is maintained through deterministic local logic while providing forensic logging and cloud-based

analytics for industrial risk management.

15. Drawing

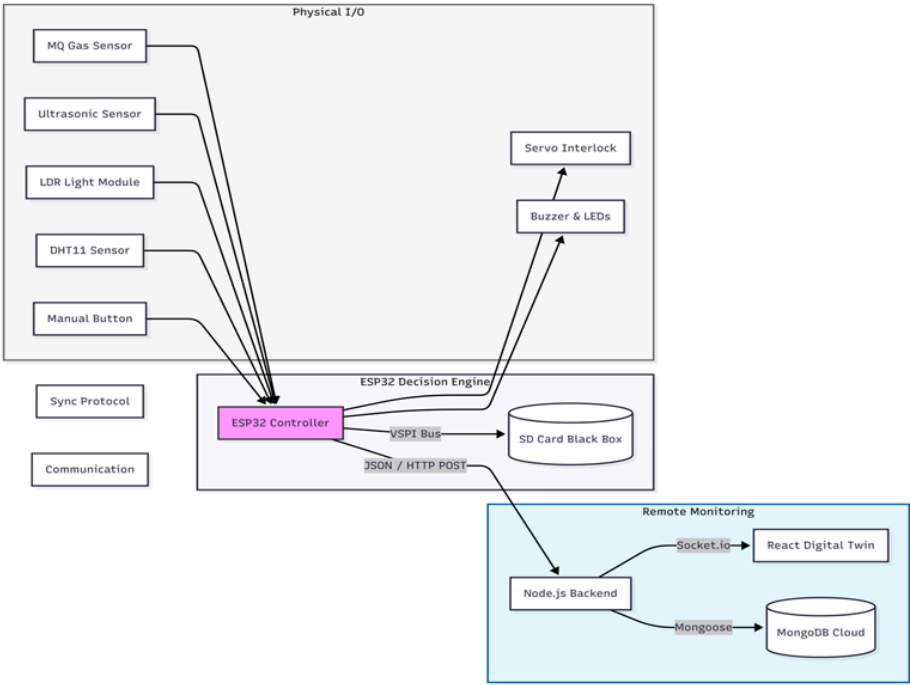


Figure 1

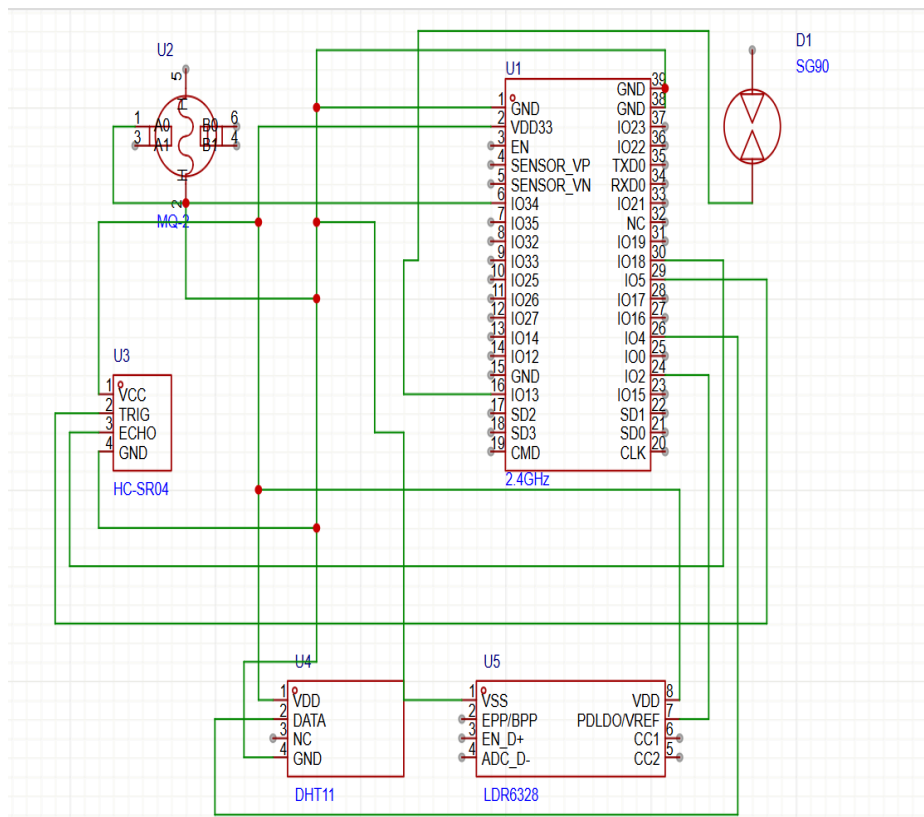


Figure 2